



OilShield — India Energy Resilience Command Center

From crisis firefighting to anticipatory decisions

AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies.

STRATEGIC INTELLIGENCE PLATFORM

ENERGY SECURITY

AI-POWERED

The Problem & Urgency

India's energy security architecture is exposed. With the vast majority of crude oil imported, any disruption in global supply corridors translates directly into economic vulnerability. The absence of intelligent, responsive tooling compounds this risk — economies without such systems take an estimated **47 extra days** to recover from oil shocks.

88%

Crude Import Dependency

Of total crude oil consumed by India is imported, leaving the economy exposed to global supply volatility.

40–45%

Hormuz Transit Share

Of India's crude imports transit the Strait of Hormuz — a single, geopolitically fragile chokepoint.

9.5

Days of Strategic Reserves

India's current Strategic Petroleum Reserve (SPR) buffer — critically insufficient for sustained disruption.

47

Extra Recovery Days

Additional days economies take to recover from oil shocks without smart, responsive decision tooling.

Solution Overview

OilShield is an integrated, single-screen command center that transforms raw supply-chain chaos into structured, auditable decisions in seconds. Built on a "**deterministic core, probabilistic edges**" principle, every output is transparent and reproducible — a critical requirement for national-level strategic planning.



Live Risk Radar

Ingests real-time news and signals, extracting structured risk scores per corridor and country with full evidence traceability.



Disruption Scenario Simulator

Projects day-by-day cascade effects across refinery run-rate, fuel price, SPR days-of-cover, and GDP under configurable assumptions.



Adaptive Procurement Recommender

Ranks alternative supply options using a transparent weighted scoring model, with plain-language rationale attached to every recommendation.



A one-click pipeline turns the chaos of a crude-oil supply shock into a structured decision in seconds.

Feature Detail — Live Risk Radar

The Live Risk Radar is the system's sensory layer — continuously ingesting raw news feeds and geopolitical signals, then converting unstructured text into structured, actionable risk intelligence. An LLM extractor (Groq/Gemini) processes incoming data, with a deterministic fallback ensuring reliability even when model outputs are ambiguous.

Signal Ingestion

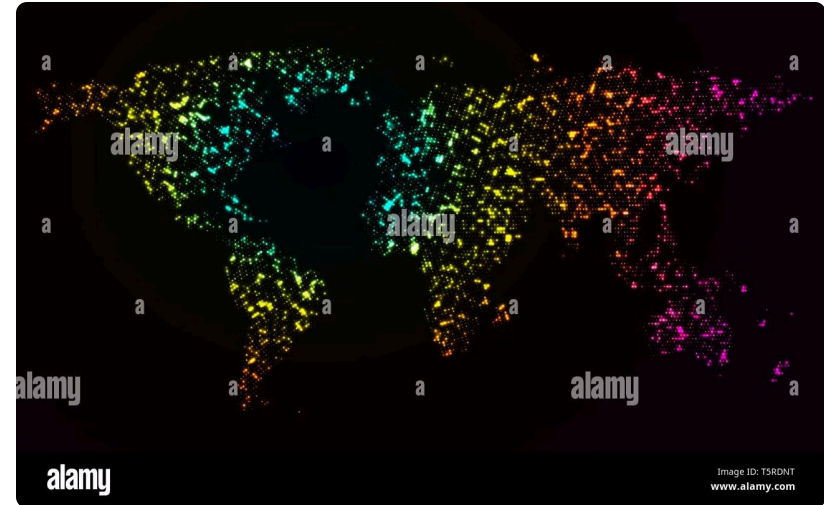
Raw news and signals ingested via LLM extraction with deterministic fallback for reliability.

Risk Aggregation

Per-corridor and per-country risk scores computed using noisy-OR style mathematical aggregation.

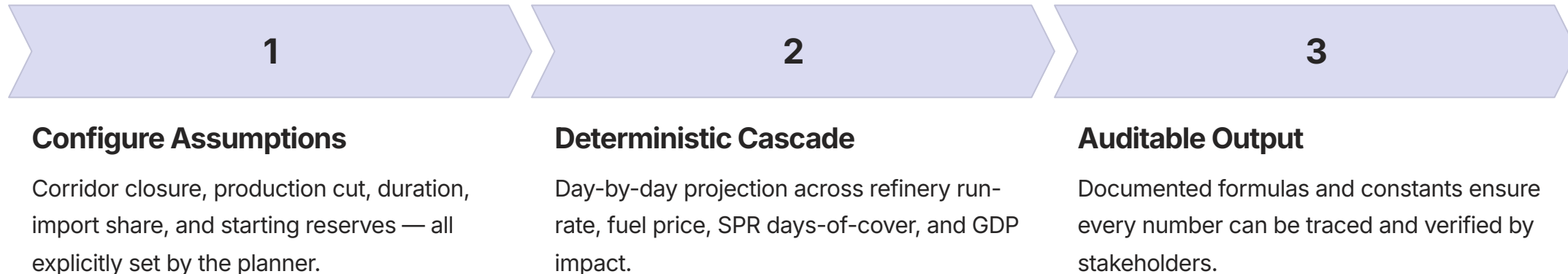
Strict Classification Bands

Low (0–33), Elevated (34–66), High (67–100) — with full evidence traceability to source.



Feature Detail — Disruption Scenario Simulator

The Simulator enables planners to move from reactive guesswork to proactive scenario planning. By configuring explicit, auditable assumptions, decision-makers can visualize the full cascade of a supply disruption before it materializes — and prepare accordingly.



 Every formula and constant used in the simulation is documented and auditable — judges and stakeholders can see exactly why a number moved.

Feature Detail — Adaptive Procurement Recommender

When a disruption is detected, the Recommender instantly evaluates alternative supply options and ranks them using a transparent, weighted scoring model. Incompatible grades are filtered out before ranking, ensuring only viable alternatives are surfaced. Every recommendation includes a plain-language rationale — no black-box decisions.

Price

Weight: **0.35**

Grade Compatibility

Weight: **0.30**

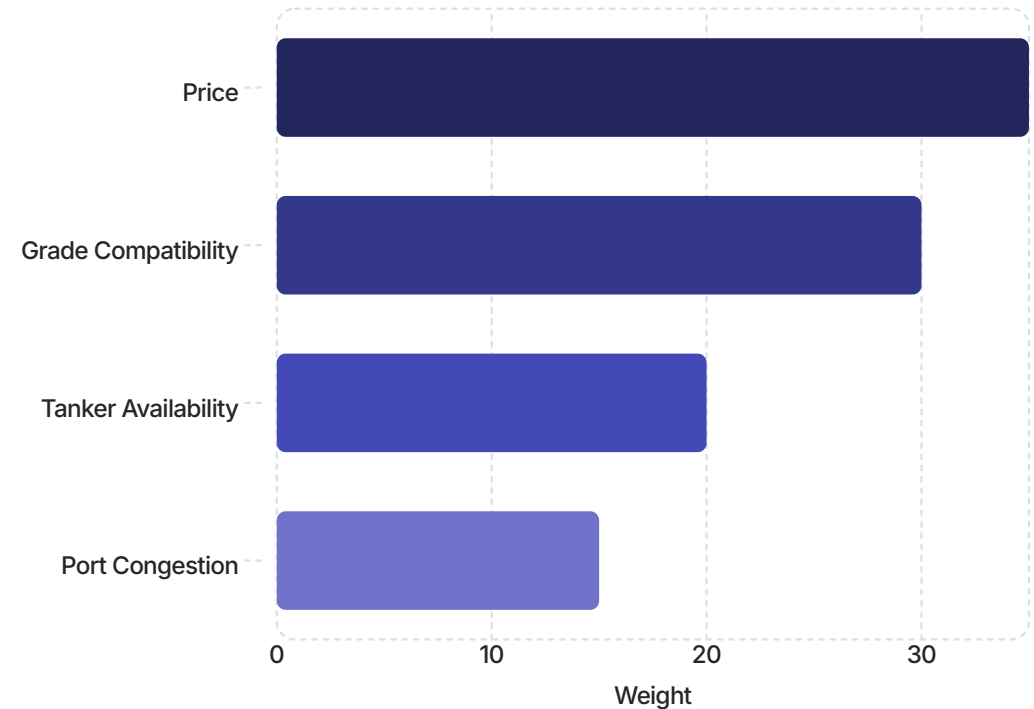
Tanker Availability

Weight: **0.20**

Port Congestion

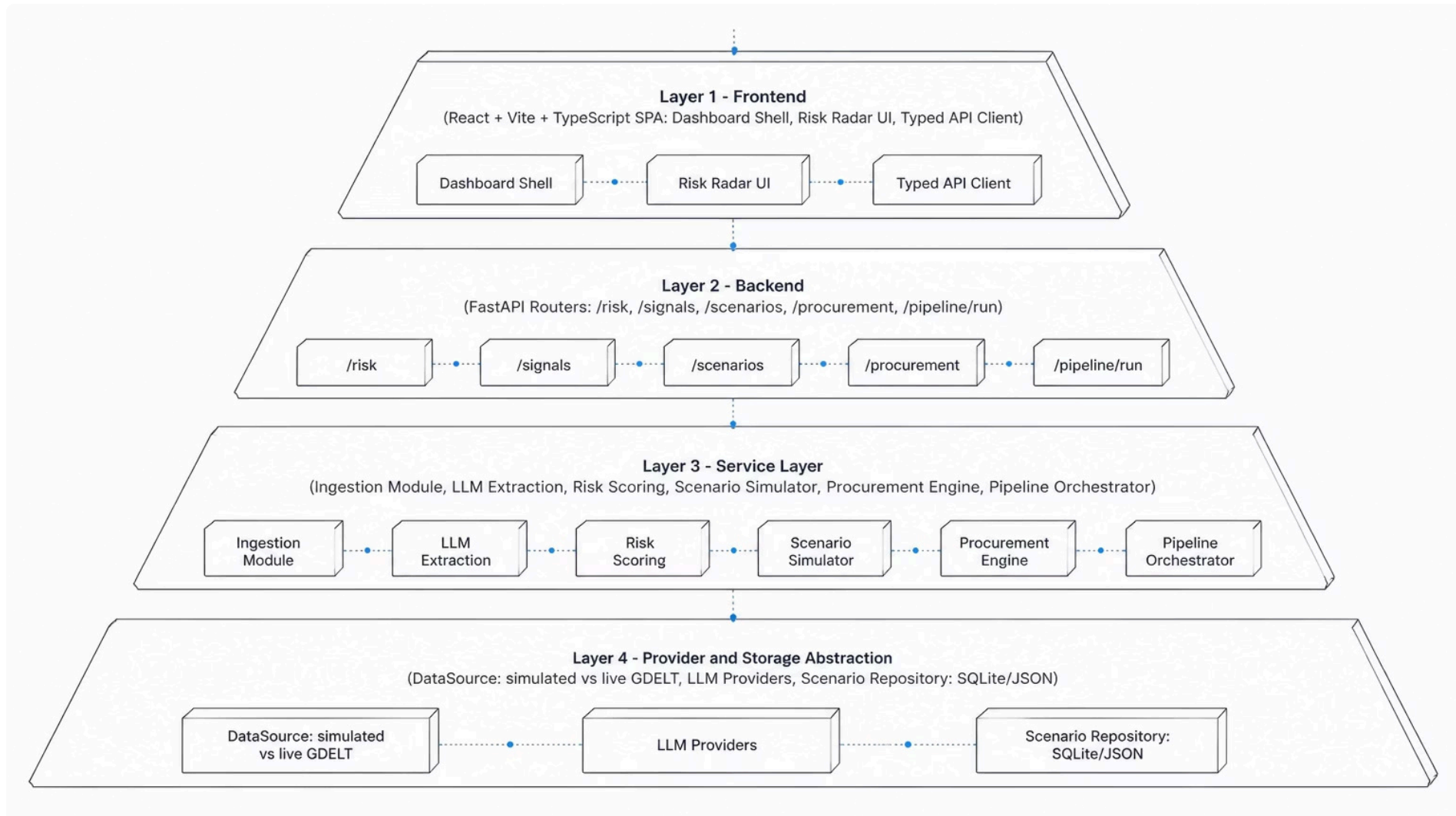
Weight: **0.15**

Factor



System Architecture — Deep Dive

OilShield's architecture is designed for clarity, modularity, and resilience. Each layer has a well-defined responsibility, enabling independent testing, replacement, and scaling. The system gracefully degrades to simulated data when live feeds are unavailable — ensuring the command center never goes dark.



The architecture follows a clean separation of concerns — from the React/TypeScript frontend through FastAPI backend routers, into the service layer where core intelligence resides, down to provider and storage abstractions that support both live and simulated data sources.

Engineering Quality & Resilience

OilShield is not just a prototype — it is engineered to production-grade standards with rigorous type safety, graceful degradation, and comprehensive testing. These engineering choices ensure the system performs reliably under real-world conditions, including network interruptions and data gaps.



End-to-End Typed Models

Full synchronization using **Pydantic** on the backend and **TypeScript** on the frontend — eliminating silent data mismatches and ensuring contract integrity across the entire stack.



Graceful Degradation

The system defaults to **offline simulated data** when live feeds are unavailable, ensuring the demo and operational environment never go dark — even during network interruptions.



Testing Rigor

46 backend tests pass across unit and integration suites, with **23 formally specified correctness properties** — providing mathematical confidence in core computation outputs.

Honest Limitations & Roadmap

CURRENT LIMITATIONS

What We Acknowledge Today

Illustrative Demo Numbers

Current outputs use illustrative figures rather than historically calibrated forecasts — limiting predictive accuracy for real deployments.

Open API Without Auth

The system currently relies on an open API endpoint without authentication — a gap that must be closed before production use.

ROADMAP

Path to Production Deployment

01

Calibrate Models

Benchmark against historical oil-shock data for forecast accuracy.

02

Live AIS & Commodity Feeds

Integrate real-time vessel tracking and commodity pricing data.

03

Push Notifications & Alerts

Set configurable alert thresholds for proactive risk notification.

04

Multi-Country Modeling

Expand scenario engine to support comparative national analysis.

05

Production Hardening

Implement RBAC, rate limiting, and migrate to PostgreSQL.

UAE-Russia Oil Trade Routes via Dubai & C

Mumbai Port



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Conclusion

From a 47-day scramble to a 15- second decision.

OilShield represents a fundamental shift in how import-dependent economies respond to energy supply shocks. By fusing live risk intelligence, deterministic scenario simulation, and transparent procurement recommendations into a single command interface, it eliminates the paralysis that currently defines crisis response.

Anticipate

Live Risk Radar surfaces threats before they become crises.

Simulate

Scenario Simulator projects full cascade impact with auditable logic.

Act

Procurement Recommender delivers ranked, rational alternatives instantly.



Strategic Imperative: For a nation that imports 88% of its crude and holds only 9.5 days of strategic reserves, the difference between a 47-day recovery and a 15-second decision is not incremental — it is existential.